Homework 6

Q1A With the sample size provided, the minimum is 102. The next smallest is 115.

That means if 102 is removed [Jackknife is LOOCV], minimum is 115. For the rest 9 that are removed, 115 is the smallest

$\hat{\theta}_{(i)} = (115, 102, 102, 102, 102, 102, 102, 102, 102)$

The equation for the Jackknife estimate is the following:

$\widehat{\theta_{jack}} = n\bar{\theta}_{(\cdot)} - (n-1)\hat{\theta}$ Where:

$\hat{\theta} = 102$

$\bar{\theta}_{\cdot} = [(115 + (9 \cdot 102))/10] = 103.3$

Plugging into the equation

$\widehat{\theta_{jack}} = 10(103.3) - 9(102)$

$= 1033 - 918$

$= \cancel{105}\ 115$

The following are the ranges for each vocal group

Voice type	Ranges (Hz)
Bass	82-110
Baritone	110-130
Tenor	130-150
Alto	160-200
Soprano	200-250

- The Jackknife estimate of 115 Hz best fits along the Baritone voice type

$X_{(1)}$ = Smallest Observation (102)
$X_{(2)}$ = 2nd Smallest observation (115)

Plugged in Jacknife equation is the following

$$\bar{\theta}_{(\cdot)} = \frac{X_{(2)} + (n-1)X_1}{n}$$

$$\hat{\theta}_{jack} = n\bar{\theta}_{\cdot} - (n-1)X_1$$

With the numbers plugged in, $X_2 = 115$, Confirming our results